

CONTACT INFORMATION	SMILE Lab 427 Richards Hall, Northeastern University 360 Huntington Avenue, Boston, MA 02115	Homepage: https://jianglin954.github.io/ E-mail: lu.jiang@northeastern.edu Phone: +1 857 206 1798
RESEARCH INTERESTS	Machine Learning, Computer Vision, Natural Language Processing, Generative Artificial Intelligence LLMs, VLMs, VLA, AI agent, Multimodal Alignment, Reinforcement Learning, World Models	
EXPERIENCE	Amazon AGI , Full-Time, Boston, USA May 2026—Now Applied Scientist Intern Manager: Dr. Chengwei Su Research Areas: GenAI, AI Agents Amazon Web Services , Full-Time, Seattle, USA May 2025—Aug 2025 Applied Scientist Intern Mentors: Dr. Hongcheng Wang , Dr. Yuanwei Wu , Dr. Ziyi Zhao Research Areas: LLMs, VLMs, AI Agent, Reinforcement Learning, Image Restoration Project: developed an agentic system for complex image restoration by fine-tuning a VLM to identify degradations, employing an LLM for restoration planning, and training a reinforcement learning agent to optimize tool execution sequences (CVPR 2026). Adobe Research , Full-Time, San Jose, USA Sep 2024—Jan 2025 Research Scientist Intern Mentors: Dr. Simon Jenni , Dr. Kushal Kafle , Dr. Jing Shi Research Areas: LLMs, VLMs, Multimodal Alignment, Retrieval, Recommendation Project I: proposed a quality-controllable search framework that trains an LLM for query refinement and integrates a VLM for text–image semantic matching, providing customers more aesthetic and higher-quality recommendations (ICLR 2026). Project II: proposed the Indra Representation Hypothesis , showing that independently trained unimodal foundation models implicitly converge to a shared relational structure of reality . Formulated representations using V-enriched Yoneda embeddings, ensuring that the resulting representations are unique, complete, and structure-preserving (NeurIPS 2025). Tencent Music Entertainment , Full-Time, Shenzhen, China Nov 2020—Apr 2021 Research Scientist Intern Mentors: Dr. Kwok-Wai Hung , Dr. Simon Lui Research Areas: Graph Learning, Hashing, Domain Adaptation, Retrieval, Recommendation Project: proposed an efficient asymmetric hashing approach for cross-modal and cross-domain search by learning an adaptive bipartite graph that models modality- and domain-level relationships, resulting in more accurate search results (TCYB 2023). EDUCATION Northeastern University , Boston, USA Jan 2023 — Now Ph.D. Candidate in Computer Engineering Advisor: Prof. Yun Raymond Fu (MAE, Fellow of ACM, AAAS, AAAI, IEEE, IAPR) INVITED TALKS ACL Lab, Northeastern University, Boston, USA 26/02/2024 Topic: Latent Graph Inference from Shallow to Deep Invited by: Prof. Sarah Ostadabbas	

Multimodal Alignment of Foundation Models

Feb 2026—Now

Collaborators: [Zhenshuo Zhang](#), Prof. [Hongyang R. Zhang](#)[Virtuoso Lab](#), Boston, USA

Project: focused on cross-sample modality attribution, proposed a structured kernel surrogate framework to evaluate its attribution quality with the linear datamodeling score, and evaluated the method on the MultiBench and MMTraCE benchmarks (submitted to **NeurIPS 2026**).

Post-Training Quantization

May 2024—Now

Collaborators: [Hailing Wang](#), Prof. [Yun Raymond Fu](#)[SMILE Lab](#), Boston, USA

Project I: proposed a PTQ method for image super-resolution, identifying that **activation outliers are crucial for color fidelity**. Introduced a dual-region quantization strategy and sensitivity-aware finetuning, achieving performance comparable to quantization-aware training without retraining while **significantly improving model efficiency (ICCV 2025)**.

Project II: provided a theoretical analysis of quantization perturbations in **MoE LLMs**, designed a **routing alignment** objective and a theoretically grounded **expert quantization strategy** to improve quantization on multiple MoE LLMs (submitted to **ICML 2026**).

LLMs for Scientific Recognition

Feb 2024–Aug 2025

Collaborators: [Yixuan Liu](#), Prof. [Albert-László Barabási](#)[Barabasi Lab](#), Boston, USA

Project I: proposed a **scale-free graph-language model** that unifies graph construction and language modeling via a **scale-free structural prior**, addressing unrealistic edge assumptions and limited supervision in prior GLMs, enabling semi-supervised learning (**ICLR 2025**).

Project II: presented a systematic evaluation of scientific recognition in LLMs and uncover persistent disparities by gender and region (**EMNLP 2025**).

1. **Jianglin Lu**, Simon Jenni, Kushal Kafle, Jing Shi, Handong Zhao, Yun Fu. [Seeing Through Words: Controlling Visual Retrieval Quality with Language Models](#), The Fourteenth International Conference on Learning Representations (**ICLR**), 2026.
2. Qihua Dong, Yang Kuo, Ju Lin, Handong Zhao, Yitian Zhang, Yizhou Wang, Huimin Zeng, **Jianglin Lu**, Yun Fu. [Ref-Adv: Exploring MLLM Visual Reasoning in Referring Expression Tasks](#), The Fourteenth International Conference on Learning Representations (**ICLR**), 2026.
3. Hailing Wang, **Jianglin Lu**, Yitian Zhang, Yun Fu. ACBQ: Adaptive Cross-Block Quantization of Large Language Models, The 64th Annual Meeting of the Association for Computational Linguistics (**ACL**), 2026
4. **Jianglin Lu**, Yuanwei Wu, Ziyi Zhao, Hongcheng Wang, Felix Jimenez, Abrar Majeedi, Yun Fu. [Restore-R1: Efficient Image Restoration Agents via Reinforcement Learning with Multimodal LLM Perceptual Feedback](#), The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) Findings, 2026.
5. **Jianglin Lu**, Hailing Wang, Kuo Yang, Yitian Zhang, Simon Jenni, Yun Fu. [The Indra Representation Hypothesis for Multimodal Alignment](#), The Fortieth Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
6. **Jianglin Lu**, Yixuan Liu, Yitian Zhang, Yun Fu. [Scale-Free Graph-Language Models](#), The Thirteenth International Conference on Learning Representations (**ICLR**), 2025.
7. Hailing Wang, **Jianglin Lu**, Yitian Zhang, Yun Fu. [Outlier-Aware Post-Training Quantization for Image Super-Resolution](#), International Conference on Computer Vision (**ICCV**) [Highlight](#), 2025.
8. **Jianglin Lu**, Hailing Wang, Yi Xu, Yizhou Wang, Kuo Yang, Yun Fu. [Representation Potentials of Foundation Models for Multimodal Alignment: A Survey](#), The 2025 Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2025. (Cited by Prof. [Yoshua Bengio](#) in the International AI Safety Report 2026)
9. Yixuan Liu, Abel Elekes, **Jianglin Lu**, Rodrigo Dorantes-Gilardi, Albert-Laszlo Barabasi. [Unequal Scientific Recognition in the Age of LLMs](#), The 2025 Conference on Empirical Methods in Natural Language Processing (**EMNLP**) Findings, 2025.
10. **Jianglin Lu**, Yi Xu, Huan Wang, Yue Bai, Yun Fu. [Latent Graph Inference with Limited Supervision](#), Thirty-Seventh Conference on Neural Information Processing Systems (**NeurIPS**), 2023.

MANUSCRIPTS UNDER REVIEW	1. Jianglin Lu, Hailing Wang, Xu Ma, Qihua Dong, Mingyuan Zhang, Yizhou Wang, Yun Fu. MUSE: A Unified Agentic Harness for MLLMs, The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026. 2. Jianglin Lu, Hailing Wang, Kuo Yang, Yi Xu, Yun Fu. Degree-Centric Structure Adaptation, 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026. 3. Jianglin Lu, Hailing Wang, Kuo Yang, Yitian Zhang, Yun Fu. Improving Multimodal Alignment for Foundation Models via Indra Representation, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026. 4. Ziniu Zhang, Zhenshuo Zhang, Jianglin Lu, Ruoxuan Xiong, Yun Fu, Hongyang R. Zhang. Designing Kernel Surrogate Models for Multimodal Attribution, The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026. 5. Hailing Wang, Jianglin Lu, Huimin Zeng, Wei Li, Lin Chen, Yun Fu. Routing-Aligned Cross-Expert Compression for Mixture-of-Experts LLMs, Forty-Third International Conference on Machine Learning (ICML), 2026. 6. Haichao Zhang, Yijiang Li, Shwai He, Tushar Nagarajan, Mingfei Chen, Jianglin Lu, Ang Li, Yun Fu. ThinkJEPA: Empowering Latent World Models with Large Vision-Language Reasoning Model, The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026.
TEACHING	Northeastern University, Boston, USA EECE7205 Fundamentals of Computer Engineering Fall 2025 EECE5642 Data Visualization (Outstanding Teaching Award) Spring 2025 DS5220 Supervised Machine Learning Summer 2023, Spring 2024
PROFESSIONAL SERVICES	Reviewer for Conference: ICML, NeurIPS, ICLR, CVPR, ICCV, ECCV, AAAI, MM, KDD, ACL, EMNLP, AISTATS Reviewer for Journal: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Transactions on Knowledge and Data Engineering (TKDE), IEEE Transactions on Image Processing (TIP), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), ACM Transactions on Knowledge Discovery from Data (TKDD), International Journal of Machine Learning and Cybernetics (IJMLC), Multimedia Tools and Applications, Expert Systems With Applications, Pattern Recognition, Neural Networks, Scientific Reports, Neurocomputing
HONORS AND AWARDS	Northeastern Outstanding PhD Teaching Award, 2025 ICCV Travel Award, 2025 ICLR Travel Award, 2025 NeurIPS Scholar Award, 2023 Northeastern PhD Network Travel Award, 2023, 2025 Second Prize of National Graduate Student Mathematical Modeling Competition, 2021 (Top 2%) China National Scholarships, 2020 (1/400)
PERSONAL NOTES	Personal Reading List Awesome Paper on Representation Alignment Awesome On-Policy Distillation Latent Graph Inference from Shallow to Deep Introduction to Vision-Language-Action Models Introduction to Vision-Language Models Introduction to Domain Adaptation Introduction to Graph Neural Networks Introduction to Learning with Noisy Labels